

INVITED TALK / ABSTRACT

Zoll families of minimal spheres and min-max theory

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Abstract

The min-max theory for the area functional on the space of embedded two-spheres in a Riemannian three-sphere defines four min-max numbers, known as the spherical area widths. We explain how the equality case of the spherical area widths characterises a class of Riemannian metrics that can be seen as higher dimensional analogues of Zoll metrics (metrics whose geodesics are all simple closed curves with the same length). This is based on joint work with Fernando Marques (Princeton) and André Neves (UChicago).

Keywords: Zoll families; minimal spheres; min-max theory; area functional; embedded two-spheres; Riemannian three-sphere; spherical area widths; Zoll metrics